

## Econometrics II

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## Problem Set 4

Due: March 6, 2026

**Question 1.** Use “dataHW4 Problem1.mat” to complete this exercise. In this problem you will work with a multinomial choice model very similar to the one from previous problem sets. Upon graduating high school, individuals have three options: work ( $Y=0$ ), attend a 2-year college ( $Y=1$ ), attend a 4-year college ( $Y=2$ ). The available student demographic characteristics ( $X_i$ ) are a constant, an indicator for whether either parent graduated from a 4-year college, and the student’s high school GPA respectively. There is one choice specific variable, price measured in thousands of dollars ( $Z_{price}$ ). Each column of  $Z_{price}$  reflects the price for the corresponding choice. In other words, the first column is the price for the work choice, the second column for the 2-year choice, and so on. Note that I assume that price for the work option is zero.

(a) Estimate a mixed logit using simulation with  $R = 500$  draws. The coefficient on price is normally distributed with mean  $\mu$  and variance 2. The unknown coefficients for the student demographic variables do not vary in the population. In other words, the only random coefficient is on price. Normalize the utility of work to zero. Report parameter estimates and SEs for the  $s$ ,  $\mu$ , and  $\sigma$ .

*Solution.*

| Table 1: Mixed Logit Estimates |           |            |
|--------------------------------|-----------|------------|
|                                | Parameter | S.E.       |
| <i>2-Yr</i>                    |           |            |
| $\beta_{0,1}$                  | -0.930576 | (0.377443) |
| $\beta_{BA,1}$                 | 0.210334  | (0.113812) |
| $\beta_{GPA,1}$                | 0.416573  | (0.122459) |
| <i>4-Yr</i>                    |           |            |
| $\beta_{0,2}$                  | -2.315272 | (1.022829) |
| $\beta_{BA,2}$                 | 0.701975  | (0.239203) |
| $\beta_{GPA,2}$                | 1.176881  | (0.295022) |
| $\mu_\gamma$                   | -0.148575 | (0.008202) |
| $\sigma_\gamma$                | 0.015835  | (1.587908) |

(b) Calculate the average marginal effect of having a parent with a BA on the likelihood of attending a 4-year college. To calculate the average marginal effect you will need to integrate out over the price coefficient, similar to what you did in estimation. Here you can use 100 draws per observation. Then bootstrap standard error.

*Solution.* Average Marginal Effect of Parent Education on 4-Year College Attendance

| AME      | Standard Error |
|----------|----------------|
| 0.111298 | 0.028921       |

(c) Re-estimate the mixed logit model from part (a) using 5-point Gaussian-Hermite quadrature to approximate the integral over the choice probabilities.

*Solution.*

Table 2: Mixed Logit Estimates (Gaussian-Hermite Quadrature)

|                 | Parameter  | S.E.          |
|-----------------|------------|---------------|
| 2-Yr            |            | $\beta_{0,1}$ |
| -0.919686       | (0.072787) |               |
| $\beta_{BA,1}$  | 0.217438   | (0.039067)    |
| $\beta_{GPA,1}$ | 0.428476   | (0.021781)    |
| 4-Yr            |            |               |
| $\beta_{0,2}$   | -2.348588  | (0.111559)    |
| $\beta_{BA,2}$  | 0.719959   | (0.040966)    |
| $\beta_{GPA,2}$ | 1.208237   | (0.02856)     |
| $\mu_\gamma$    | -0.155445  | (0.006060)    |
| $\sigma_\gamma$ | 0.038863   | (0.093009)    |

**Question 2.** Use “dataHW4 Problem2.mat” to complete this exercise. In this problem you will work with panel data to estimate a logit model that allows for discrete unobserved heterogeneity. During the pandemic working from home became the norm, and many people continue to work from home. Over the course of six months, you collect data from women between the ages of 30 and 40 who are working full-time. Each month you observe whether the woman was working at home full time ( $Y=1$ ). The six columns of  $Y$  reflect her response for each month. For each respondent you also observe  $X_i$  containing the number of children at home and years of schooling (in addition to a constant). You also observe distance from the physical workplace in miles ( $Z_{dist}$ ). Note that  $X_i$  and distance are fixed for each woman over the 6 months.

(a) You suspect that in addition to children and schooling, women likely differ in their tastes for working at home. Assume that there are two unobserved types of women, one type prefers the office, the other prefers to work from home, all else equal. Type is assumed to be uncorrelated with  $X_i$  or distance. Estimate a mixed logit model for the decision to work from home that allows for these two types of women. Note that this means there will be two additional parameters to estimate beyond the coefficients associated with  $X_i$  and distance. You need to estimate the proportion of Type 2 women, and the additional preference these women have for working at home. Report coefficients and SEs.

*Solution.*

Table 3: Mixed Logit Estimates (Two Types of Women)/EM Algorithm

|                               | Parameter | S.E.       |
|-------------------------------|-----------|------------|
| <i>Likelihood Estimates</i>   |           |            |
| $\beta_0$                     | -0.468813 | (0.181709) |
| $\beta_{Children}$            | 0.511097  | (0.011169) |
| $\beta_{Schooling}$           | -0.204464 | (0.007455) |
| $\gamma$                      | 0.052225  | (0.000165) |
| $\delta$                      | 1.98194   | (0.068723) |
| $\pi_2$                       | 0.680251  | (0.020931) |
| <i>EM Algorithm Estimates</i> |           |            |
| $\beta_0$                     | -0.468813 | —          |
| $\beta_{Children}$            | 0.511097  | —          |
| $\beta_{Schooling}$           | -0.204464 | — $\gamma$ |
| 0.052225                      | —         |            |
| $\delta$                      | 1.98194   | —          |
| $\pi_2$                       | 0.680251  | —          |

The standard errors from the EM algorithm could be obtained through the numerical Hessian, taking the inverse of the Hessian at the estimated values provides the covariance matrix. From there we can take the root of the diagonal elements. If this is too computationally taxing, bootstrapped standard errors can be obtained by running the EM on  $B$  iterations of the data and calculating the standard deviation of the parameter estimates across the  $B$  iterations.